1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist's path indicates PROS1 is expressed in tumors and is associated with Ki-67, implying PROS1-related signaling may drive or reflect proliferation; the key question is under what microenvironmental conditions PROS1 associates with increased proliferation and worse outcome. 2) Tier 1 paths used: (a) Tumor -> PROS1 (PROS1 expressed in tumor tissue) and (b) PROS1 -> Ki-67 association (links PROS1 to proliferation). I built on these because they form the simplest mechanistic backbone connecting a tumor-expressed ligand to a canonical proliferation readout. 3) Tier 2/supporting evidence: The ultrasound → gene expression non-effect supports that measured PROS1/MKI67 levels reflect biology rather than imaging artifact, and the organ/microenvironment context (bladder in the raw path) motivates testing how stromal context modulates PROS1's effect. 4) Why OS chosen: Overall survival is the most robust, clinically consequential endpoint available in TCGA and is the outcome most plausibly impacted by sustained increases in proliferation (MKI67) driven by microenvironment-modulated PROS1 signaling. 5) Deliberate exclusions: I excluded additional molecular actors (e.g., specific TAM receptor gene names, coagulation cascade genes) to keep the hypothesis sharply focused on the PROS1→proliferation axis and an imaging-derived microenvironmental modifier; those factors are reserved for secondary or follow-up analyses because they were not enumerated as anchor genes in the provided Ontologist output?

Working Hypothesis

PROS1 expression in breast tumor tissue is positively associated with MKI67 (Ki-67) levels, and this association is amplified in tumors with low stromal-to-tumor ratio (low TSR). Consequently, high PROS1 with low TSR predicts poorer overall survival (OS) compared with low PROS1 or high-TSR contexts. The effect is mediated by microenvironmental modulation of PROS1 signaling pathways that promote proliferation, resulting in a significant PROS1 × TSR interaction on MKI67 and a corresponding prognostic interaction on OS.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	PROS1, MKI67
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PROS1: PROS1 is expressed in tumors and is associated with Ki-67, implying PROS1-related signaling may drive or reflect proliferation; the key question is under what microenvironmental conditions PROS1 associates with increased proliferation and worse outcome.

MKI67: MKI67 levels reflect biology rather than imaging artifact, and the organ/microenvironment context (bladder in the raw path) motivates testing how stromal context modulates PROS1's effect.

3. Clinically Testable Predictions

Prediction 1: PROS1 is a ligand that can engage TAM receptor signaling, potentially promoting proliferative programs in tumor cells;

Prediction 2: MKI67 is a robust readout of cell proliferation and is prognostically relevant in breast cancer;

Prediction 3: TSR captures stromal content relative to tumor cells and reflects the microenvironmental context that can modulate signaling and tumor behavior. Integrating PROS1 and MKI67 with TSR allows testing a context-dependent PROS1-driven proliferation hypothesis and its impact on OS.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions